

GROUP 6 · GENERATION GHANA & KENYA DATA ANALYTICS PROGRAMME

Workforce Analytics

Analyst Report: A Data-Driven Diagnosis of Talent, Performance & Retention Risk

Prepared for Roblox Africa Operations · Ghana Hub Leadership

Roblox HRMS · 30,000-Employee Workforce Study · 8 Departments

Data Pipeline: Excel Cleaning → MySQL → Power BI → Executive Reporting

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LIVE INTERACTIVE DASHBOARD**Power BI Service**

All visuals in this report are static exports. For the live, filterable version with slicers for department, year and role

open the dashboard directly in Power BI:

app.powerbi.com/links/rJjy2d9ucw

Note: this is a workspace share link. It requires a Power BI account within our organizational tenant. Contact Group 6 if access is needed.

Executive Summary

Roblox Africa Operations has positioned Ghana Hub as a strategic recruitment base for engineering, data, customer support and operations talent, and invested significantly in the past five years into development, health benefits and compensation to drive that strategy. This report addresses the questions raised by those investments - does the organisation get value for money in terms of placement, does education predict placement and pay, is the workforce productive and healthy - by analysing two data sources delivered for this capstone: the seven executive SQL queries run directly on the MySQL warehouse, and the four Power BI dashboards built from the same warehouse. Where the two sources address similar questions, this report cross-checks them rather than presenting them separately, and notes the few areas where they conflict. Together, they paint a picture of an organisation that delivers consistently good performance from its 30,000 employees across eight departments: average performance per department falls within a 3.58–3.61 range, project delivery at scale (822,492 completed), and no department combines above-average cost with below-average performance. There are three areas in which the workforce analytics raise concerns.

- Revenue efficiency shows wide variation while performance is stable - the best department generates \$1.75 of revenue per \$1 of cost, the worst only \$1.38, despite both delivering within 0.03 of performance.
- Workforce risk is widespread and evenly distributed - insurance/wellbeing risk touches 65–68% of employees in all departments, and \$208.7M of total compensation falls to at-risk staff.
- Training investment does not follow efficiency - Human Resources delivers the best revenue-cost ratio (1.75×) with the shortest training time (401 hours), and Finance enjoys the most training (2,788 hours) for the second-best ratio (1.74×).

The workforce is productive and healthy, but the organisation sees broadly similar financial returns per department despite wide variation in cost, and has a wellbeing/insurance risk exposure that touches all departments.

Data & Methodology

Sources

Two sources feed into this report: the seven relational tables described in the Roblox HRMS Data Dictionary, queried directly via the seven executive SQL scripts, and the four Power BI dashboards built from the same tables. The dashboards provide a few high-level workforce metrics (gender split, average age, role distribution, attrition) not found in the SQL exports, and the report labels all figures by source. Section 8 reconciles the two sources where they overlap.

The upstream cleaning that feeds the MySQL warehouse (de-duplication, type/format standardisation, key mapping in Excel) is documented separately in the project's cleaning workbook and is not repeated here - this report operates on the cleaned data and focuses on analysis.

Query design

The employee table sits at the centre of a star schema, connected one-to-many to employee_performance and education, and via department_code to department and department_performance. One-to-one relationships see employee join to finance (compensation details) and health (wellbeing risk).

Seven queries were run against this schema to address executive questions. Each query that touches employee_performance or education aggregates those tables to an employee-level view before joining to employee, avoiding inflated headcount or compensation figures at the 1:many join.

Query	Business question
1 · Workforce, Cost & Wellbeing Overview	What does the workforce look like by department, and what compensation / wellbeing exposure sits behind it?
2 · Department Performance & Financial Efficiency	Which departments convert workforce investment into stronger performance and financial output?
3 · Education, Placement & Compensation	How does education background relate to placement and pay?
4 · Employee Cost vs Productivity	Which employees combine compensation investment with measurable productivity?
5 · Workforce Risk & Cost Exposure	Where are wellbeing risks concentrated, and what compensation exposure sits behind them?
6 · Executive Department Scorecard	One comparable view of scale, cost, performance, productivity and risk, by department.
7 · Executive Outlier Analysis	Which departments run above-average cost with below-average performance?

Note on scope: departmental revenue and cost figures below reflect the latest recorded year per department (as selected in Query 6), consistent with the \$23.34M revenue / \$14.68M cost totals already shown on the Power BI dashboard. Where a figure is drawn from a cumulative, all-years view instead, this is stated explicitly.

Findings

1. Workforce Composition, Cost & Wellbeing (Query 1)

Headcount is deliberately even: every department falls within a 3,699–3,819 range, and no function dominates the org chart. Compensation follows the same pattern - average basic salary is broadly similar across departments (~\$8.7K–\$8.8K) - but wellbeing risk is not evenly distributed across the workforce in absolute terms: it is uniformly high. Between 65% and 68% of employees in all departments carry a wellbeing/insurance risk flag, and total compensation across all departments reaches \$312.5M (\$262.9M basic salary + \$49.5M allowances).

Department	Headcount	Avg Salary	Total Compensation	Wellbeing Risk %
Customer Support	3,819	\$8,750	\$39.74M	68.2%
Data & Analytics	3,810	\$8,799	\$39.82M	67.1%
Human Resources	3,784	\$8,835	\$39.72M	66.7%
Marketing	3,718	\$8,789	\$38.78M	66.2%
Engineering	3,699	\$8,811	\$38.67M	66.7%
Product Management	3,725	\$8,728	\$38.64M	65.0%
Finance	3,727	\$8,704	\$38.59M	66.3%
Operations	3,717	\$8,704	\$38.53M	67.2%

A definitional note for stakeholders: only 10,077 of 30,000 employees (33.6%) carry an employee_status of "Active" in the source data - implying a 66.4% inactive rate that is roughly double the 32.9% attrition figure already reported on the dashboard. This suggests employee_status and the dashboard's attrition calculation are not measuring the same thing, and should be reconciled with the source system owner before either number is used in a board-level document.

Power BI cross-check (Dashboard 01 - Workforce Composition): the SQL exports above do not break out gender, age or role, so these three data points come from the Power BI dashboard. Gender is close to evenly split (15,007 male vs. 14,993 female), average age across the workforce is 40.98 years, roughly two-thirds of employees hold a Bachelor's or Master's degree, and the four broad role bands - Executive, Consultant, Analyst, Developer - are each staffed to roughly the same size, mirroring the flat, deliberate headcount balance already visible by department in the SQL table above.

2. Department Performance & Financial Efficiency (Query 2 / Scorecard)

Performance itself barely varies between departments: it falls within a 3.58–3.61 range, with Operations at the low end and Engineering/Data & Analytics at the high end. Financial efficiency - revenue per dollar of cost - shows a far wider spread, from 1.38× for Customer Support to 1.75× for Human Resources. In other words, two departments can perform almost identically well and still generate revenue at very different efficiency relative to their cost.

Department	Performance	Revenue	Cost	Revenue : Cost
Human Resources	3.59	\$3.80M	\$2.17M	1.75×
Finance	3.59	\$4.37M	\$2.51M	1.74×
Engineering	3.61	\$3.42M	\$2.08M	1.64×

Department	Performance	Revenue	Cost	Revenue : Cost
Operations	3.58	\$3.52M	\$2.18M	1.62×
Marketing	3.59	\$2.16M	\$1.40M	1.55×
Data & Analytics	3.61	\$1.23M	\$0.86M	1.43×
Product Management	3.59	\$2.98M	\$2.13M	1.40×
Customer Support	3.60	\$1.86M	\$1.35M	1.38×

Read together: Finance is the largest revenue-generating department (\$4.37M) and also enjoys the second-best revenue-cost ratio (1.74×), a combination that speaks to its strategic importance to the business. Customer Support and Product Management both combine below-average revenue with above-average cost, and thus have the greatest potential to improve their financial performance.

3. Education, Placement & Compensation (Query 3)

Education shows little relationship to pay: across all placements, average total compensation falls within a narrow \$10,409 for Master's degree holders, \$10,387–\$10,471 for Diploma holders and \$10,520 range for Bachelor's and Doctoral degrees, respectively. Business Administration dominates as a field of study, with 8,553 employees, far exceeding the 3,800–4,030 range seen in Computer Science, Data Analytics, Information Technology, Economics and Engineering.

By department, Human Resources enjoys the best average compensation across roles (\$10,525) and Product Management the lowest (\$10,359), and this follows the expected pattern of a higher average for Executives (\$166 (1.6%) spread that is dwarfed by the department-level revenue-to-cost gap above. By position, Managers (\$10,475) and Consultants (\$10,457) sit modestly ahead of Assistants (\$10,389), a normal and expected hierarchy premium.

The practical implication for leadership: pay in this organisation is dictated more by role and department than by education attainment. A Doctoral-qualified Analyst and a Diploma-qualified Analyst in the same department are likely to be compensated similarly - worth noting whether that reflects an intentional de-emphasis of education as a pay criterion or an unexpected outcome.

Power BI cross-check (Dashboard 02 - Education & Compensation): the dashboard independently reports Master's degree holders as the single largest education segment (10.6K employees), just edging out Bachelor's (10.0K). This compares very closely to the weighted figures derived from the SQL exports (10,566 Master's vs. 9,867 Bachelor's, Query 3). The dashboard and the report also agree on department-level cost: the dashboard's highest and lowest total workforce cost - Data & Analytics (\$39.8M) and Operations (\$38.5M) - match the SQL Query 1 figures in Section 1 to within \$20K. This is a useful integrity check: two independently built views of the same warehouse agree on these figures.

4. Employee Cost vs Productivity - the top-performer profile (Query 4)

This query's export captures the top 1,000 employees by performance score - the best 3.3% of the workforce, not the full population - so it is most useful read as a definition of "good" rather than an analysis of the workforce as

a whole. Within that top 1,000, 894 employees (89.4%) are flagged High productivity / strong attendance and the remaining 106 (10.6%) fall into Monitor, meaning even among top performers roughly one in nine shows a softer attendance or consistency signal worth a manager check-in.

Customer Support (92.8%) and Engineering (91.0%) have the highest share of "High productivity / strong attendance" employees within this top tier; Operations (85.5%) and Marketing (86.4%) have the lowest, which is consistent with Operations also showing the highest attrition risk elsewhere in the deck's Dashboard 04 findings.

5. Workforce Risk & Cost Exposure (Query 5)

Risk exposure is remarkably flat across the organisation: every department falls within a narrow 65.0–68.2% band for insurance/wellbeing risk, and average medical leave balance is essentially identical everywhere (14.76–15.19 days). This is a structural, organisation-wide exposure rather than a localised departmental issue, and it has a direct financial impact: \$23.34M revenue / \$0__312.5M compensation base - sits with employees currently flagged at wellbeing/insurance risk.

Department	Insurance Risk %	Comp. Exposed to Risk	Avg Medical Leave Bal.
Customer Support	68.2%	\$27.29M	15.1 days
Data & Analytics	67.1%	\$26.81M	15.0 days
Human Resources	66.7%	\$26.44M	14.8 days
Operations	67.2%	\$25.96M	14.9 days
Engineering	66.7%	\$25.87M	15.1 days
Marketing	66.2%	\$25.58M	15.0 days
Finance	66.4%	\$25.57M	15.2 days
Product Management	65.0%	\$25.18M	15.0 days

Customer Support carries both the highest insurance-risk share (68.2%) and the lowest revenue-to-cost ratio (1.38×) of any department - the only function where a wellbeing exposure and a financial-efficiency gap point in the same direction, which makes it the single highest-priority department for a combined HR and finance review.

Power BI cross-check (Dashboard 04 - Workforce Risk & Wellbeing): the dashboard reports 9,920 employees with expired insurance coverage - a narrower, more specific cut than the SQL wellbeing/insurance risk flag used in Section 1 and the table above (66.7% org-wide), which appears to capture a broader set of risk conditions rather than expired coverage alone. The two figures are not directly comparable and should not be quoted interchangeably, but both point the same direction: a material share of the workforce carries unresolved wellbeing or coverage exposure. The dashboard also reports Operations as the single highest-attrition department at 35.1%, well above the org-wide 32.9% rate, despite Operations sitting mid-pack on performance (3.58) - direct, independent support for Recommendation 1 below.

6. Executive Department Scorecard - full comparison (Query 6)

Bringing scale, cost, performance, productivity and risk together in one view confirms the core tension in this dataset: departments are nearly indistinguishable on performance (3.58–3.61) and risk exposure (65–68%), but pull apart sharply on financial efficiency (1.38×–1.75×) and training investment (401–2,788 hours).

Department	Headcount	Performance	Rev:Cost	Training Hrs	Risk %
Finance	3,727	3.59	1.74×	2,788	66.4%
Operations	3,717	3.58	1.62×	1,431	67.2%
Human Resources	3,784	3.59	1.75×	401	66.7%
Product Management	3,725	3.59	1.40×	2,636	65.0%
Engineering	3,699	3.61	1.64×	1,102	66.7%
Marketing	3,718	3.59	1.55×	748	66.2%
Customer Support	3,819	3.60	1.38×	1,464	68.2%
Data & Analytics	3,810	3.61	1.43×	839	67.1%

There is no visible relationship between training hours and either performance or revenue efficiency in this table. Human Resources achieves the best revenue-cost ratio in the organisation on the least training investment by a wide margin (401 hours, next-lowest is Marketing at 748), while Finance and Product Management post the two highest training totals for a top-tier and a bottom-tier efficiency ratio, respectively. This corroborates the dashboard's existing "training spend doesn't clearly predict performance" finding and extends it: it doesn't clearly predict financial efficiency either.

7. Executive Outlier Analysis (Query 7)

Organisation-wide average compensation is \$10,398 and average performance is 3.54 (employee-level average, computed across all annual performance records). Five of eight departments - Human Resources, Engineering, Data & Analytics, Marketing and Customer Support - run above-average compensation. None of the eight combine above-average cost with below-average performance, so there is no department in active "priority review" territory on this specific test.

Department	Avg Compensation	Avg Performance	Flag
Human Resources	\$10,497	3.59	Higher compensation exposure
Engineering	\$10,454	3.61	Higher compensation exposure
Data & Analytics	\$10,452	3.61	Higher compensation exposure
Marketing	\$10,429	3.59	Higher compensation exposure
Customer Support	\$10,406	3.60	Higher compensation exposure
Product Management	\$10,372	3.59	Within benchmark
Operations	\$10,366	3.58	Within benchmark
Finance	\$10,355	3.59	Within benchmark

The reassurance here is real - no department is overpaying for underperformance. The caution is subtler: because performance barely varies (a 0.03 spread) while five departments are paying above the organisation average, higher compensation in this dataset is not clearly buying differentiated performance. It may be buying something else - retention, seniority mix, or role composition - that this dataset doesn't yet isolate, and is worth a follow-up query joining compensation outliers to tenure and role level.

8. Triangulating SQL and Power BI

Combining the two sources strengthens some findings and surfaces one open question that neither source resolves alone.

- Agrees - cost by department: Data & Analytics has the highest total workforce cost and Operations the lowest, in both the SQL export (Section 1) and the Power BI dashboard, to within \$20K - strong evidence the two are drawing from a consistent, well-reconciled warehouse.
- Agrees - education mix: Master's-degree headcount as the largest education segment, ahead of Bachelor's, in both the SQL weighted count (Query 3) and the dashboard - a second independent confirmation of pipeline integrity.
- Agrees - Operations is the outlier to watch: Operations shows the weakest link between effort and outcome in both sources: the SQL scorecard's lowest performance (3.58) and the dashboard's highest single-department attrition (35.1%) are describing the same department from two different angles.
- Needs reconciliation - status, attrition & risk definitions: employee_status "Active" share (33.6%, SQL) and the dashboard's attrition rate (32.9%) are not the same metric and should not be quoted as if they were - see Data Quality note below. Likewise the dashboard's expired-insurance count (9,920 employees) and the SQL wellbeing/insurance risk flag (66.7%) appear to use different eligibility rules and need a single owner to define one company-standard risk metric.

Data Quality & Definitional Notes

- employee_status vs. attrition rate: Only 33.6% of employees carry status "Active" (Query 1/6), while the dashboard reports 32.9% attrition - these numbers describe different things and should not be used interchangeably without a definitions note from the source-system owner.
- Query 2 export naming: the file exported under the "Department Performance & Financial Efficiency" label contains the Education, Placement & Compensation columns (Query 3's output) rather than Query 2's year-by-year revenue-to-cost columns. This report used the Executive Department Scorecard (Query 6) as the authoritative source for Query 2's revenue-to-cost metric, and treated the mislabeled file as Query 3 content - worth relabeling the export before this is filed as a permanent record.
- Employee Cost vs Productivity sample size: this export is capped at the top 1,000 employees by performance score, not the full 30,000 - findings in Section 4 describe the top-performing segment only and should not be read as an organisation-wide productivity split.
- Scope of financial figures: department-level revenue and cost in Sections 2 and 6 reflect each department's latest recorded year, matching the dashboard's single-year view (\$23.34M revenue /

\$1__14.68M cost org-wide). An earlier dashboard export briefly showed a cumulative all-years figure (\$23.34M revenue / \$2__65.51M); both are correct for their respective year filters, but only one should be labelled "current year" performance in the final deck.

Recommendations

1 · Retention - investigate Operations

Operations combines steady 3.58 performance with the highest attrition already flagged on Dashboard 04 (35.1%) and the lowest "high productivity / strong attendance" share among top performers (85.5%). Investigate workload, compensation structure and working conditions before the department's output quality erodes.

2 · Financial efficiency - prioritise Customer Support and Product Management

These two departments underperform since their revenue is below average while the costs are above average (1.38× and 1.40× revenue-cost ratio, respectively), which makes them the two lowest-performing departments. In addition, the Customer Support department has the highest wellbeing-risk share (68.2%), so the manager should take a closer look at their process before thinking about any cost increases.

3 · Compensation strategy - reconcile the outlier flags

There are five departments with compensation above the organisational average, and apparently, they are not underperforming compared to others. To understand the situation, perform an additional analysis cross-referencing compensation outliers with tenure, seniority mix, and role level, as the difference might be justified by the structural difference between the groups.

4 · Wellbeing & insurance - treat as a systemic, org-wide programme

Since insurance/wellbeing risk is at 65-68% across departments, this is not an issue affecting a certain team, but a problem systematically affecting all employees, which requires a top-down approach to resolution. Allocating funds to address this issue across all departments would be more efficient than tackling it one team at a time, as the affected compensation range would be \$208.7M either way.

5 · Data governance - fix the definitional gaps before board reporting

To ensure there is nothing critical missed before presenting this report to the board, it would be important to reconcile the employee_status field with the attrition rate, relabel the query 2 export ("Query2_export.csv"), and timestamp the files indicating whether revenue/cost figures represent the figures for the latest year or the cumulative value.

Appendix: Organisation-Wide Metrics at a Glance

Metric	Value
Total employees	30,000

Metric	Value
Departments	8
Total compensation	\$312.49M
basic salary	\$262.94M
allowances	\$49.54M
Revenue generated	\$23.34M
Total department cost	\$14.68M
Revenue : cost ratio	1.59
Projects completed	822,492
Organisational performance score	3.60
Employees at wellbeing/insurance risk	20,010 / 66.7%
Total compensation exposed	\$208.71M
Average compensation (employee)	\$10,398
Average performance (employee)	3.54